2023BCSE008

Lab: 05 Date:- 02|04|2026

Q1. Write a Java Program to demonstrate all the three types (Single, Hierarchical, & Multilevel) Inheritance.

Lab_05 > Main1.java > Main1

```java
//1. Write a Java Program to demonstrate all the three types (Single, Hierarchical, & Multilevel) Inheritance.
package Lab_05;
class A {
    void showA() {
        System.out.println(x: "Class A");
    }
}

class B extends A { // single
    void showB() {
        System.out.println(x: "Class B");
    }
}

class C extends B { // multilevel
    void showC() {
        System.out.println(x: "Class C");
    }
}

class D extends A { // hierarchical
    void showD() {
        System.out.println(x: "Class D");
    }
}

public class Main1 {
    Run | Debug
    public static void main(String[] args) {
        B b = new B();
        b.showA();
        b.showB();

        C c = new C();
        c.showA();
        c.showB();
        c.showC();

        D d = new D();
        d.showA();
        d.showD();
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_05 % javac -d . Main.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main
Class A
Class B
Class A
Class B
Class C
Class A
Class D
```

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Q2. Write a Java program to create a class Person with:

• Private variable: *name*

• Public methods:

- *setName(String n)*
- *getName()*

Create an object and access the variable using getter and setter methods.

Lab_05 > Main2.java > Main2

```java
package Lab_05;
class Person {
    private String name;

    public void setName(String n) {
        name = n;
    }

    public String getName() {
        return name;
    }
}

public class Main2 {
    Run | Debug
    public static void main(String[] args) {
        Person p = new Person();
        p.setName(n: "Annu");
        System.out.println(p.getName());
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_05 % javac Main2.java
gauravmathur@Annus-Mac Lab_05 % javac -d . Main2.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main2
Annu
```

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Q3. Write a Java program to demonstrate protected access modifier:

• Create a class Parent with protected variable value
• Create a subclass Child those accesses and displays the value
Create an object of Child and display the result.

Lab_05 > Main3.java > Child > display()

```java
package Lab_05;
class Parent {
    protected int value = 100;
}

class Child extends Parent {
    void display() {
        System.out.println(value);
    }
}

public class Main3 {
    Run | Debug
    public static void main(String[] args) {
        Child c = new Child();
        c.display();
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac Lab_05 % javac Main3.java
gauravmathur@Annus-Mac Lab_05 % javac -d . Main3.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main3
100
```

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Q4. Write a Java program to access a class (say Scanner or some userdefined) from a package using **fully qualified name** without using import statement.

Lab_05 > Main4.java > Main4

```java
package Lab_05;

public class Main4 {
    Run | Debug
    public static void main(String[] args) {
        java.util.Scanner sc = new java.util.Scanner(System.in);
        System.out.println(x: "Helloo");
        sc.close();
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac Lab_05 % javac Main4.java
gauravmathur@Annus-Mac Lab_05 % javac -d . Main4.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main4
Hello
```

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Q 5. Write a Java program to create a class ***DatabaseConnection*** that implements the Singleton Design Pattern. The Singleton pattern means that only one object of a class can be created from outside the class.

Lab_05 > Main5.java > Main5

```java
package Lab_05;
class DatabaseConnection {
    private static DatabaseConnection instance;

    private DatabaseConnection() {
        System.out.println(x: "Connection created");
    }

    public static DatabaseConnection getInstance() {
        if (instance == null) {
            instance = new DatabaseConnection();
        }
        return instance;
    }
}

public class Main5 {
    Run | Debug
    public static void main(String[] args) {
        DatabaseConnection obj1 = DatabaseConnection.getInstance();
        DatabaseConnection obj2 = DatabaseConnection.getInstance();

        System.out.println(obj1 == obj2);
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_05 % javac Main5.java
gauravmathur@Annus-Mac Lab_05 % javac -d . Main5.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main5
true
```

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Q6. Write a Java program to create a class Printer that implements the Doubleton Design Pattern. The Doubleton pattern means that only two objects of a class can be created from outside the class.

Lab_05 > Main6.java > Main6

```java
package Lab_05;
class Printer {
    private static Printer obj1 = new Printer();
    private static Printer obj2 = new Printer();
    private static int count = 0;

    private Printer() {
    }

    public static Printer getInstance() {
        if (count == 0) {
            count++;
            return obj1;
        } else if (count == 1) {
            count++;
            return obj2;
        } else {
            return null;
        }
    }
}

public class Main6 {
    Run | Debug
    public static void main(String[] args) {
        Printer p1 = Printer.getInstance();
        Printer p2 = Printer.getInstance();
        Printer p3 = Printer.getInstance();

        System.out.println(p1);
        System.out.println(p2);
        System.out.println(p3);
    }
}
```

OUTPUT:-

```
gauravmathur@Annus-Mac Lab_05 % javac Main6.java
gauravmathur@Annus-Mac Lab_05 % javac -d . Main6.java
gauravmathur@Annus-Mac Lab_05 % java Lab_05.Main6
Lab_05.Printer@7ad041f3
Lab_05.Printer@251a69d7
null
```